Team 155

Problem B

Two Points to the Goal: A Noise-Conditioned Mixture Density Network for Probabilistic Penalty Kick Prediction

Summary

From two early observations of a penalty kick, we predict the ball's crossing of the goal plane and quantify its uncertainty. We compare a constant-acceleration baseline, adaptive Bayesian inference and a noise-conditioned mixture density network (MDN) based on simulations with gravity, quadratic drag and Magnus lift.

At an assumed coordinate noise standard deviation of 5 mm, the calibrated MDN gives a mode of $(Y, Z) = (2.62, 2.15)$ m and joint 50% and 95% region areas of 0.177 and 0.775 m^2. Conditional on airborne crossing, the ball centre has a 97.3% probability of lying inside the goal rectangle. The Bayesian reference gives 96.3%.

On 5,036 independently simulated airborne crossings, the MDN's median point error is 0.290 m at 5 mm. Across four noise levels, the calibrated inner and outer regions cover 47.6–49.1% and 94.5–95.2% of simulated endpoints. A 3 $\mathrm{m\,s^{-1}}$ crosswind lowers inner-region coverage from 48.7% to 40.9% on matched flights. Tracked flights provide case-level checks of the physical model.

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1 Introduction

1.1 Problem Restatement

Take the penalty spot as the origin and the goal as the rectangle $|Y| < 3.66\,\mathrm{m}$, $0 < Z < 2.44\,\mathrm{m}$ in the plane $x = 11\,\mathrm{m}$ [1]. From $\mathbf{r}(0) = (0, 0, 0.11)\,\mathrm{m}$ and two later observations, the contest asks for the most likely crossing point and its uncertainty. The stated case gives [2]

$$\mathbf{r}(0.05) = (1.50, 0.45, 0.44)\,\mathrm{m}, \qquad \mathbf{r}(0.10) = (2.96, 0.86, 0.74)\,\mathrm{m}. \tag{1}$$

Unknown spin and aerodynamic forces make extrapolation from two points ambiguous. With a fixed start, 5 mm independent coordinate errors give a second-difference acceleration uncertainty of $4.47\,\mathrm{m\,s^{-2}}$. For the stated kick, the same noise gives a median radial deviation of about 0.34 m under quadratic extrapolation (Appendix A).

1.2 Background and Approach

Wind-tunnel tests of non-spinning balls found drag coefficients near 0.43 and 0.15 on either side of the drag crisis, at Reynolds numbers around $2.2\text{–}3.0 \times 10^5$ [3]. A later ball comparison reached 3.3×10^5 [4]. Free-kick fits give $C_D = 0.25\text{–}0.30$. In the same study ten free kicks had speeds of $23.0\text{–}28.3\,\mathrm{m\,s^{-1}}$ at 0.1 s [7]. Trajectory studies find lift coefficients of order 0.2–0.3 under some conditions and estimate aerodynamic curves from tracked flights [5, 6]. These results inform our choices of prior bounds and lift law.

We compare Bayesian inversion and an MDN [19] under a shared simulator. Independent simulations assess predictive coverage, and ISSIA-3D and DFL data provide external checks on the physical trajectories [14, 15, 16].

2 Materials and Methods

2.1 Assumptions and Approximations

Forecasts use the known start, fixed observation times and specified coordinate noise σ. Endpoint distributions condition on reaching the goal plane before ground contact, denoted by E. We report the density mode for the stated case and use the mean for point-error scoring.

A1. Ball. A rigid sphere has mass 0.430 kg, radius 0.110 m and cross-section $A = \pi R^2$ [1]. Buoyancy is omitted (Section 2.2).

A2. Air. The main model uses still air at $\rho = 1.20\,\mathrm{kg\,m^{-3}}$.

A3. Drag. Each flight has a constant $C_D \sim U[0.15, 0.30]$, which includes reported free-kick and supercritical wind-tunnel estimates [7, 3].

A4. Lift. Transverse spin determines lift. The slope and cap in Eq. (5) are chosen to match the measured order of magnitude [5, 3].

A5. Spin. We hold spin fixed, following the short-flight approximation discussed by Team 358 [8]. Section 3.5 tests a 0.5 s decay time.

A6. Measurement. The start is exact. The two observations have independent Gaussian coordinate errors with common standard deviation $\sigma \in \{2, 5, 10, 15\}$ mm in evaluation.

A7. Prior. The independent uniform bounds in Table 1 describe plausible simulated launches.

Table 1: Prior bounds for the launch parameters (independent uniforms).

Parameter	Lower	Upper
v_x ($\mathrm{m\,s^{-1}}$)	18	35
v_y ($\mathrm{m\,s^{-1}}$)	-12	12
v_z ($\mathrm{m\,s^{-1}}$)	0.5	12
$\omega_x, \omega_y, \omega_z$ ($\mathrm{rad\,s^{-1}}$)	-60	60
C_D	0.15	0.30

2.2 Physical Model

Let $\mathbf{u} = \mathbf{v} - \mathbf{w}$ be the velocity relative to air, $\mathbf{g} = -g\hat{\mathbf{z}}$ with $g = 9.81\,\mathrm{m\,s^{-2}}$, and $V = 4\pi R^3/3$. The force balance combines weight, quadratic drag, transverse lift and Archimedean buoyancy [9, 10, 11, 12].

$$m\dot{\mathbf{v}} = m\mathbf{g} - \tfrac{1}{2}\rho A C_D |\mathbf{u}|\mathbf{u} + \tfrac{1}{2}\rho A C_L |\mathbf{u}|^2 \hat{\mathbf{n}} - \rho V \mathbf{g}. \tag{2}$$

Buoyancy is 0.066 N upward, or 1.6% of the ball's weight. Omitting it gives

$$\dot{\mathbf{r}} = \mathbf{v}, \tag{3}$$

$$\dot{\mathbf{v}} = \mathbf{g} - \frac{\rho A C_D}{2m}|\mathbf{u}|\mathbf{u} + \frac{\rho A C_L}{2m}|\mathbf{u}|^2\hat{\mathbf{n}}. \tag{4}$$

We use a three-dimensional spin vector $\boldsymbol{\omega}$ and define

$$\hat{\mathbf{n}} = \begin{cases} \dfrac{\boldsymbol{\omega}\times\mathbf{u}}{|\boldsymbol{\omega}\times\mathbf{u}|}, & |\boldsymbol{\omega}\times\mathbf{u}| > 0, \\ \mathbf{0}, & \text{otherwise}, \end{cases} \qquad C_L = \min\!\left(1.4\,\frac{R\,|\boldsymbol{\omega}\times\mathbf{u}|}{|\mathbf{u}|^2},\, 0.30\right). \tag{5}$$

Spin about $+\hat{\mathbf{z}}$ gives the special case $\hat{\mathbf{n}} \propto \hat{\mathbf{z}} \times \mathbf{u}$. Lift vanishes when the cross product is zero. Bray and Kerwin give a related vector model for free kicks [7].

2.3 Numerical Solution and Event Detection

The synthetic simulator uses fourth-order Runge–Kutta (RK4), a 2 ms step and a 1.5 s horizon. Linear interpolation locates $x = 11\,\mathrm{m}$ and $z = R$, retaining goal-plane crossings before contact. Halving the step on the first fresh launch (seed 155107) changes the endpoint by less than 0.001 mm. The ISSIA model-form check uses `solve_ivp`. Tracked-flight prefix forecasts use RK4 with a maximum 5 ms step.

2.4 Bayesian Inference of Launch Conditions

Let $\boldsymbol{\theta} = (\mathbf{v}_0, \boldsymbol{\omega}, C_D)$ and let $\mathbf{y}_1, \mathbf{y}_2$ be the two observed positions. With the start fixed and independent Gaussian errors,

$$p(\mathbf{y}_1, \mathbf{y}_2 \mid \boldsymbol{\theta}) = \prod_{i=1}^{2} \frac{1}{(2\pi\sigma^2)^{3/2}} \exp\!\left[-\frac{\|\mathbf{y}_i - \mathbf{r}(t_i;\boldsymbol{\theta})\|^2}{2\sigma^2}\right]. \tag{6}$$

Four-stage adaptive importance sampling combines a 20% prior component with a Gaussian proposal. The first three stages use possibly widened likelihoods to adapt the proposal by weighted moments. The fourth stage alone supplies the posterior at the specified σ, with weights proportional to likelihood times $p(\boldsymbol{\theta})/q(\boldsymbol{\theta})$. We use 40,000 draws per stage for the case and 6,000 for the paired test, retrying with 18,000 when conditional effective sample size (ESS) is below 100. A weighted Gaussian kernel estimate on a 160×160 grid gives the mode and joint highest-density regions (HDRs) [24].

2.5 Noise-Conditioned Mixture Density Network

The network estimates $p(Y, Z \mid \mathbf{y}_1, \mathbf{y}_2, \sigma, E)$. Its eleven inputs are midpoint velocity and second-difference acceleration (three components each), speed, tangential and normal aerodynamic acceleration features, constant-acceleration time to the plane, and σ. We standardise using training data and fill an undefined time feature with zero in standardised coordinates.

Three 128-unit hidden layers with LayerNorm and GELU produce $K = 3$ bivariate Gaussian components [21, 22].

$$p(Y, Z \mid \mathbf{y}_1, \mathbf{y}_2, \sigma, E) = \sum_{k=1}^{K} \pi_k \mathcal{N}((Y, Z); \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)\,. \tag{7}$$

Each component has a weight logit, two means, two scales and one correlation. The 18 outputs define softmax weights and covariances $\boldsymbol{\Sigma}_k = \mathbf{L}_k \mathbf{L}_k^\top$, with

$$\mathbf{L}_k = \begin{pmatrix} s_{k1} & 0 \\ \rho_k s_{k2} & s_{k2}\sqrt{1-\rho_k^2} \end{pmatrix}, \qquad s_{kj} = \operatorname{softplus}(r_{kj}) + 10^{-3}, \qquad \rho_k = 0.995 \tanh(r_{k\rho}), \tag{8}$$

Here r denotes a raw output and the scales refer to standardised endpoint coordinates. The network predicts absolute (Y, Z) and minimises negative log-likelihood (NLL) using Adam, an initial learning rate of 10^{-3} with cosine decay, batch size 256 and 100 epochs [13]. Each epoch draws a new $\sigma \sim U[1, 20]$ mm per flight and perturbs the six observed coordinates independently at that scale.

2.6 Data Splits, Comparisons and Calibration

Of 50,000 flights generated with seed 42, 41,733 reached the plane in flight. Before adding noise, we split these into 33,367 training, 4,179 validation and 4,187 development test flights. For P1, validation was divided into 2,090 model-selection and 2,089 calibration flights. After freezing model and calibration, we generated 6,000 new flights (seed 155107) and scored their 5,036 airborne crossings. Comparisons use matched observation perturbations.

The A–E ablation fixes the basic network and split while comparing raw coordinates (A, B), 16 physics features (C, D) and coordinates plus the baseline endpoint (E). A and C predict absolute coordinates. B, D and E predict residuals, skip undefined residuals in training and set invalid baseline endpoints to zero in testing. We retrained PI-ResMDN on the common simulator. Legacy MDN and PI-ResMDN selected epochs on the original validation set.

We screened feature count and mixture size at two SiLU layers, depth at fixed SiLU, then activation at the selected depth [23]. Selection uses four-noise-level mean validation NLL across seeds 11, 22 and 33. We prefer the simpler candidate when its NLL disadvantage lies within one paired seed standard deviation and its mean coverage gap increases by at most 1.5 percentage points.

We report Euclidean point errors, NLL in physical coordinates, HDR coverage and area, and the energy score in loss form (lower is better) [25]. For each seed and noise level, calibration fits an NLL-based scale multiplier, then adjusts HDR thresholds using density ranks estimated from 512 predictive draws per flight [26]. Scale fitting changes the density. Rank adjustment changes the regions. Both steps use the same calibration fold. Fitting uses discrete ranks, whereas evaluation uses interpolated density quantiles. We assess coverage empirically under this protocol.

2.7 Independent Reproduction

A second machine passed 14 unit tests, recomputed 256 sampled trajectories and reproduced all six arrays of the 50,000-flight dataset. It also reproduced four measured cases and 11 screened segments to 10^{-9} absolute or relative tolerance. The P1 training protocol and the data and checkpoint hashes were verified separately against the frozen records. Computations use NumPy, SciPy, PyTorch and Matplotlib [27, 28, 29, 30].

3 Results

3.1 Answer for the Stated Kick

The constant-acceleration case gives $T = 0.405\,\text{s}$ and $(Y, Z) = (2.495, 1.920)\,\text{m}$ (Appendix A). At an assumed $\sigma = 5\,\text{mm}$, the calibrated MDN (seed 11) gives mode $(2.62, 2.15)\,\text{m}$, mean $(2.70, 1.97)\,\text{m}$, joint 50% and 95% areas 0.177 and $0.775\,\text{m}^2$, and conditional in-frame probability 97.3%. The Bayesian reference gives mode $(2.650, 2.052)\,\text{m}$, areas 0.219 and $0.809\,\text{m}^2$ and 96.3% (ESS 2,893).

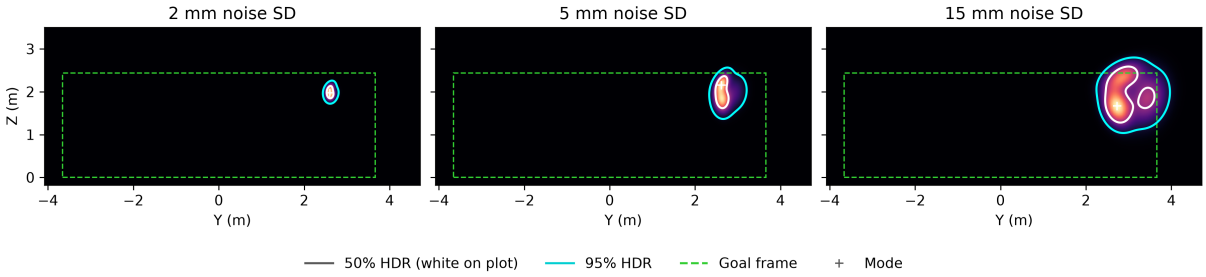


Figure 1: Scale-adjusted MDN densities for the stated kick. White and cyan mark the calibrated joint 50% and 95% contours, the dashed rectangle marks the goal, and the cross marks the mode.

Table 2: Calibrated seed-11 MDN case results. In-frame probabilities condition on airborne crossing and refer to the ball centre.

Noise (mm)	Mode Y (m)	Mode Z (m)	50% area (m^2)	95% area (m^2)	In-frame
2	2.60	1.98	0.035	0.151	$> 99.9\%$
5	2.62	2.15	0.177	0.775	97.3%
15	2.73	1.67	0.819	2.370	80.8%

At 5 mm, the MDN and physics posterior have similar 5th-to-95th percentile widths in Y (0.61 and 0.60 m) and Z (both 0.83 m). They assign 2.7% and 3.7% above the crossbar, accounting for almost all their difference in in-frame probability.

In Fig. 2, draws A and B fit the early observations within 4 mm RMSE and cross at almost the same Y, but differ in height by about 0.29 m, with opposite vertical components of $\boldsymbol{\omega} \times \mathbf{u}$. Draw C has weaker leftward lift and reaches farther right.

At fixed $Y = 2.62\,\text{m}$ the MDN has two local maxima, while all three seeds have a single peak in the marginal Z density. The Bayesian slice varies with smoothing bandwidth. The trajectories explain the spread in height. The local peak count is sensitive to the density representation (Appendix B).

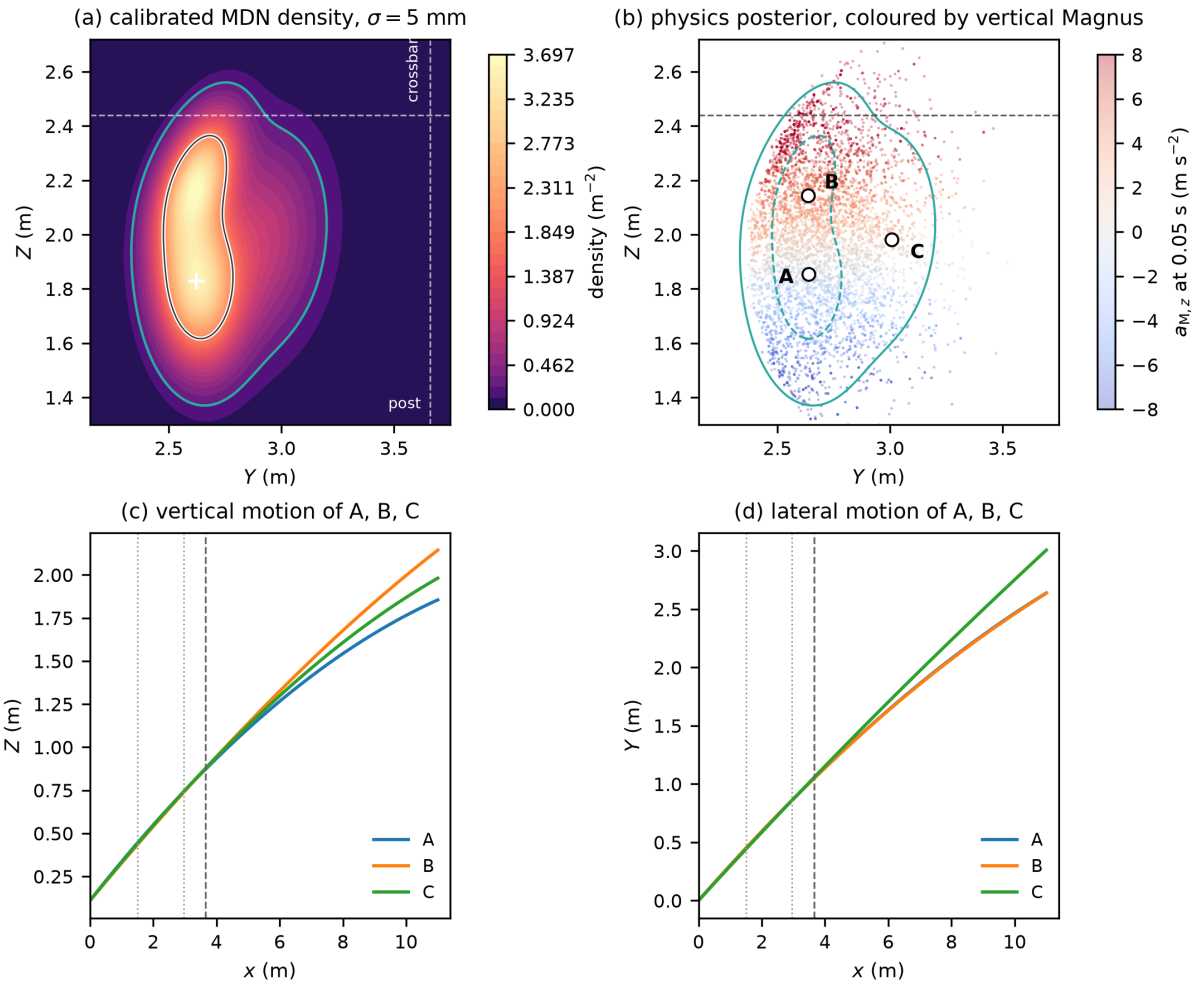


Figure 2: Physical interpretation at $\sigma = 5\,\text{mm}$. (a) MDN density and contours. (b) Weighted-resampled physics endpoints coloured by vertical Magnus acceleration evaluated at the observed midpoint velocity. (c, d) Representative trajectories from Table 3.

Table 3: Representative posterior draws. RMSE pools the six observed coordinates. Magnus accelerations (in $\text{m}\,\text{s}^{-2}$) are evaluated along each draw at 0.05 s. Responsibilities condition on the draw's endpoint.

Draw	RMSE (mm)	Crossing (Y, Z) (m)	C_D	$a_{\text{M},y}$	$a_{\text{M},z}$	Responsibility
A	3.25	(2.64, 1.86)	0.269	−9.62	−1.22	(0.08, 0.89, 0.02)
B	3.93	(2.64, 2.14)	0.263	−10.57	+5.34	(0.06, 0.24, 0.71)
C	4.97	(3.01, 1.98)	0.264	−3.84	+1.48	(0.99, 0.01, 0.00)

3.2 Feature, Target and Architecture Comparisons

The original simulators differed by 0.240 m RMS in shared airborne endpoints, so we compare PI-ResMDN after retraining it on the common simulator. Physics features improve direct prediction (C versus A in Table 4). Model D leads by NLL at 2 mm (-1.709 versus -1.682 for the final MDN), but reaches 9.88 at 15 mm. That score includes the zero-endpoint fallback. At this noise level the baseline fails on 233 development flights and 288 fresh flights.

The 11-feature model scores -0.0613 versus -0.0556 for 16 features. Switching from three components to one raises NLL to 0.1205. At fixed SiLU, the three- versus four-layer gap (0.0061) lies within the paired seed variation (0.0078). At the selected depth, GELU gives -0.1006 versus -0.0898, -0.0722 and -0.0643 for SiLU, ReLU and Tanh. This is a stepwise comparison at width 128.

Removing within-component correlation, fixing the learning rate and fixing training noise give validation NLLs of -0.0245, 0.0298 and 0.0010, respectively.

Table 4: Model comparison at 5 mm on 5,036 fresh crossings. Metrics average three seeds except for the legacy checkpoint. Only the final row uses calibration.

Model	Median (m)	90th percentile (m)	NLL
Legacy MDN	0.324	0.675	0.176
A: coordinates, direct	0.309	0.665	0.049
B: coordinates, residual	0.306	0.660	0.120
C: physical features, direct	0.293	0.643	-0.182
D: physical features, residual	0.290	0.648	0.189
E: coordinates and baseline, residual	0.292	0.650	-0.048
PI-ResMDN, canonical retraining	0.433	1.086	1.089
Calibrated final MDN	0.290	0.644	$\mathbf{-0.219}$

3.3 Blind-Test Accuracy and Calibration

At 5–15 mm, the final MDN has the lowest NLL among the compared networks, including before calibration. At 5 mm its point error nearly ties D and is within 3 mm of C. The constant-acceleration baseline has a 0.439 m median error over its 5,030 finite predictions.

Table 5: Final MDN on 5,036 independently simulated crossings, averaged over three seeds. Area and NLL are post-calibration.

Noise (mm)	Median (m)	NLL	50% pre/post	95% pre/post	95% area (m^2)
2	0.130	-1.682	49.9 / 49.1	94.8 / 94.5	0.318
5	0.290	-0.219	48.2 / 47.8	95.6 / 94.9	1.246
10	0.466	0.623	47.2 / 48.8	96.4 / 95.2	2.373
15	0.566	0.956	45.3 / 47.6	96.9 / 94.5	2.804

Raw and final inner-region coverage are below 50% at all four noise levels (Table 6). Scaling alone changes coverage by at most 0.4 percentage points. Across all three seeds, the fitted rank thresholds shrink the inner regions at 2 and 5 mm and expand them at 10 and 15 mm. Blind-test coverage falls in the first two cases and rises in the latter two. At 15 mm the two steps together raise inner coverage from 45.3% to 47.6% and lower outer coverage from 96.9% to 94.5%, shrinking the outer area from 3.162 to 2.804 m^2.

Table 6: Blind-test coverage of the nominal 50% joint region by calibration step, in percent, averaged over three training seeds.

Noise (mm)	Raw	Scale only	Scale and rank
2	49.94	50.06	49.12
5	48.25	48.31	47.79
10	47.22	47.37	48.82
15	45.27	44.94	47.63

For conditional in-frame probability, expected calibration errors are 0.0041, 0.0093, 0.0079 and 0.0089 in increasing noise order. The Brier scores are 0.0179, 0.0384, 0.0579 and 0.0671. The observed in-frame fraction among simulated airborne crossings is 43.5%.

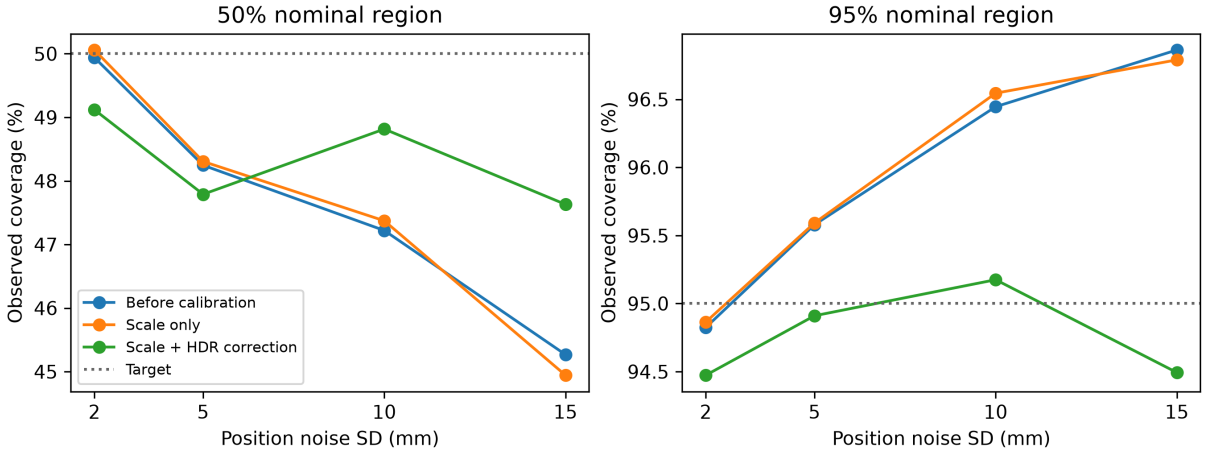


Figure 3: Region coverage on the independent simulation test. Dotted lines show nominal levels.

3.4 Paired Comparison with the Physics Posterior

Of 32 fresh flights sampled at 5 mm, 30 pass the conditional ESS threshold after retry. Median errors are 0.379 m (constant acceleration), 0.314 m (Bayesian), 0.324 m (legacy MDN) and 0.340 m (final MDN, seed 11). The Bayesian and final MDN median energy scores are 0.210 and 0.211. Median elapsed times on the same CPU are 2.04 s and 1.02 ms. Both include inference, drawing or resampling 4,096 predictive endpoints, and scoring. The small, ESS-filtered subset limits the accuracy ranking.

3.5 Distribution Shift

We hold the model and calibrator fixed and reuse launch states and observation noise at 5 mm. On flights airborne in both simulations, a 3 m s^{-1} crosswind (5,001 pairs) lowers 50% coverage from 48.7% to 40.9% and 95% coverage from 94.3% to 92.6%. Spin decay with a 0.5 s time constant (5,003 pairs) lowers inner coverage from 48.2% to 36.3% even as median error falls from 0.293 to 0.263 m. The perturbations change the crossing status of 70 and 94 of the 6,000 launches, respectively.

3.6 Sensitivity to the Drag Prior

For the stated kick, raising the upper bound of C_D from 0.30 to 0.45 shifts the posterior mean by about 2 cm. Mean 50% and 95% areas change by +1.1% and −0.5%, respectively, less than the variation across the three seeds.

Table 7: Sensitivity of the case posterior to the upper limit of the drag prior ($\sigma = 5$ mm, three seeds, 40,000 samples per stage). Ranges are minima and maxima over seeds; means are averages over seeds.

C_D prior	50% area (m^2)	95% area (m^2)	Mean (Y, Z) (m)
$U[0.15, 0.30]$	0.213 (0.205–0.219)	0.806 (0.789–0.821)	(2.696, 1.988)
$U[0.15, 0.45]$	0.215 (0.210–0.222)	0.802 (0.792–0.809)	(2.689, 1.971)

3.7 Model-Form Check on Tracked Arcs

We fit the first six frames of ISSIA-3D segments 608–622 and 1144–1172, sampled at 25 fps, and score the remaining frames [14, 15]. Selection required at least two cameras, eight consecutive frames, height of at least 0.30 m and median reprojection error below 10 pixels per frame. Absolute 3D accuracy and action labels were unverified for these segments.

Drag reduces held-out RMSE in both arcs. Adding an effective lift component in the vertical plane gives the lowest errors (Table 8). The fitted drag and lift have large uncertainties and may absorb tracking or model error.

Table 8: Held-out RMSE and last-frame error, in metres, for two ISSIA arcs. Initial conditions and any free aerodynamic coefficients use the first six frames.

Arc	Quantity	Gravity	$C_D = 0.22$	$C_D = 0.45$	Drag + lift
608–622	Forecast RMSE	0.697	0.489	0.469	**0.188**
	Final error	1.243	0.863	0.811	**0.308**
1144–1172	Forecast RMSE	2.179	1.546	1.584	**1.084**
	Final error	4.169	2.876	2.942	**2.119**

3.8 Prefix Forecasts on Tracked Flights

We fit the first three DFL positions at 0, 0.04 and 0.08 s and forecast the held-out trajectory in local coordinates [16]. Both kicks resulted in goals. The start frames are 128955 and 176244 (events 18232100000995 and 18242100001386). With 15 mm assumed noise, unknown spin improves one crossing estimate and slightly worsens the other (Table 9).

Unknown spin raises held-out RMSE from 0.338 to 0.366 m on ISSIA frames 2972–2991. For 22 held-out planar positions digitised from Barber and Carré's Fig. 5, RMSE rises from 2.266 to 2.301 m at an assumed 67 mm digitisation scale [17].

Of 60 DFL flight candidates, 11 passed motion and event screening. At both 15 and 30 mm, unknown spin lowers held-out RMSE in 7 of 11 segments, with median paired improvements of 0.027 and 0.011 m. The screen fits a quadratic to the full segment, including later frames, and favours smooth arcs. These segments lack goal-plane labels.

Table 9: Two DFL penalties with seven scored positions before each reference crossing. Both trajectory RMSE and crossing-point error are in metres. The crossing forecasts condition on airborne arrival at each observed goal plane.

	Trajectory RMSE		Crossing error	
Case	Zero spin	Unknown spin	Zero spin	Unknown spin
J03WOH	0.347	0.292	0.645	0.543
J03WR9	0.309	0.305	0.542	0.553

4 Discussion

The main gain is in forecast density. Similar point errors can coexist with different NLL and coverage, especially under wind or spin decay. The Bayesian sampler exposes assumptions and supplies a physical reference. The MDN amortises inference, and a browser implementation illustrates the effect of changing observations and assumed noise.

The forecast applies to the specified observation schedule, noise and launch prior. The prior excludes launches below $18\,\mathrm{m\,s^{-1}}$ and assigns zero mass to exact zero spin. An unconditional geometric in-frame forecast also requires $P(E \mid \mathbf{y}_1, \mathbf{y}_2, \sigma)$. The goal test uses the ball centre and omits frame contact and goalkeeper intervention. Measured examples test individual trajectories. Real-penalty coverage remains to be measured, and the PFF release provides a possible source for further cases [18].

A further observation at 0.15 s with the same 5 mm noise reduces the local estimate of median crossing spread from 0.255 to 0.103 m under the simulated launch prior (Appendix A.3). This measurement design adds 50 ms before the forecast can be made.

5 Conclusion and Outlook

Two early positions support a crossing forecast under the stated physical and measurement assumptions. At 5 mm the calibrated MDN gives mode $(2.62, 2.15)$ m and a 97.3% conditional ball-centre in-frame probability. Independent simulation supports the outer region more consistently than the inner one, whose coverage drops under physical shifts. Next steps are high-rate 3D tracking with independent timing and noise estimates, a held-out set spanning misses and ground contact, and a separate model for airborne arrival.

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A Constant-Acceleration Baseline and Noise Propagation

A.1 Baseline from the Three Given Positions

For the positions in Eq. (1) and $\Delta t = 0.05\,\mathrm{s}$, the midpoint velocity and acceleration are

$$\mathbf{v}(\Delta t) = \frac{\mathbf{r}_2 - \mathbf{r}_0}{2\Delta t} = (29.6, 8.6, 6.3)\,\mathrm{m\,s^{-1}}, \tag{9}$$

$$\mathbf{a} = \frac{\mathbf{r}_2 - 2\mathbf{r}_1 + \mathbf{r}_0}{\Delta t^2} = (-16, -16, -12)\,\mathrm{m\,s^{-2}}. \tag{10}$$

Writing $\tau = T - \Delta t$, solve $-8\tau^2 + 29.6\tau - 9.5 = 0$ and take the earlier forward root, $\tau = 0.355\,\mathrm{s}$. Substitution gives $T = 0.405\,\mathrm{s}$ and

$$(Y, Z) = (0.45 + 8.6\tau - 8\tau^2,\ 0.44 + 6.3\tau - 6\tau^2) = (2.495, 1.920)\,\mathrm{m}. \tag{11}$$

The benchmark baseline selects the earliest finite quadratic root after 0.10 s, with a linear branch when a_x is near zero. Among 5,036 reference-airborne flights, 0, 6, 111 and 288 predictions are invalid at 2, 5, 10 and 15 mm, respectively.

A.2 Propagation of Position Noise to the Endpoint

With an exact start and two independent coordinate errors of variance σ^2, the second difference has error $\varepsilon_2 - 2\varepsilon_1$ and component variance $5\sigma^2$ [20]. Thus

$$\sigma_a = \frac{\sqrt{5}\,\sigma}{\Delta t^2} = 4.47\,\mathrm{m\,s^{-2}} \quad \text{at } \sigma = 5\,\mathrm{mm}. \tag{12}$$

An independently noisy start would raise the component variance to $6\sigma^2$.

For a fixed crossing time T, let $n = T/\Delta t$. In the quadratic interpolant through the exact start and the two observations, the noisy positions have Lagrange weights $L_1 = -n(n-2)$ and $L_2 = n(n-1)/2$. The per-axis standard deviation at fixed T is $\Lambda(n)\sigma$, where

$$\Lambda(n) = \sqrt{L_1^2 + L_2^2}, \qquad e_{50}^{\mathrm{fixed}} = \sqrt{2\ln 2}\,\Lambda(n)\sigma. \tag{13}$$

Here e_{50}^{fixed} is the median radial deviation in (Y, Z) for isotropic Gaussian errors. The stated kick has $n = 8.10$ and $\Lambda = 57.2$, giving 0.337 m at 5 mm.

We also propagate the observations through 150,000 quadratic fits per noise level, solving for the crossing time in each draw. This diagnostic requires a positive midpoint-relative root, $T \le 1.5\,\mathrm{s}$ and $Z > R$ at the plane. At 5 mm its median radial deviation from the unperturbed quadratic crossing is 0.339 m.

Table 10: Constant-acceleration propagation of observation errors. Intervals contain the 5th to 95th percentiles of retained draws.

σ (mm)	Discarded	s.d. T (ms)	s.d. Y (m)	s.d. Z (m)	Y 5–95% (m)	Z 5–95% (m)
2	0	4.8	0.115	0.115	2.31–2.68	1.73–2.11
5	0	12.3	0.290	0.289	2.02–2.97	1.45–2.40
15	5,955	41.6	0.933	0.871	1.02–4.04	0.62–3.48

The spreads in Table 10 describe perturbations of the stated case. Table 5 reports Euclidean prediction errors across simulated flights.

A.3 Measurement Design and Local Crossing Spread

We analyse the first 4,000 launches of the fresh simulation set, retaining 3,332 airborne crossings. Let $\mathbf{h}(\boldsymbol{\theta})$ be the observed positions and $\mathbf{q}(\boldsymbol{\theta}) = (Y, Z)$ the crossing. We compute their Jacobians J_h and J_q by centred differences about each simulated launch. Approximating each uniform prior by a Gaussian of the same variance $(\theta_{\max} - \theta_{\min})^2/12$ gives the local crossing covariance

$$\Sigma_q = J_q(\sigma^{-2} J_h^\top J_h + \Sigma_\theta^{-1})^{-1} J_q^\top, \tag{14}$$

where Σ_θ is the diagonal prior covariance. We estimate each flight's median radial spread from 64 draws of $\mathcal{N}(\mathbf{0}, \Sigma_q)$ and report the median across flights in Table 11. The third-position variant adds a reading at 0.15 s with independent noise of the same standard deviation. These estimates describe local uncertainty under the simulated dynamics and the approximate prior.

Table 11: Local median radial crossing spread in metres for two and three noisy positions, on the same 3,332 simulated flights.

σ (mm)	0.05, 0.10 s	0.05, 0.10, 0.15 s
2	0.119	0.045
5	0.255	0.103
10	0.386	0.188
15	0.443	0.251

For a comparison of acceleration scales, let s_a be the root-mean-square of the lateral and vertical standard deviations of launch aerodynamic acceleration over these flights. Here $s_a = 7.70\,\mathrm{m\,s^{-2}}$. The ratio $\mathcal{R} = \sigma_a/s_a$, using Eq. (12), is 0.23, 0.58, 1.16 and 1.74 at 2, 5, 10 and 15 mm. The scales match at $\sigma = \Delta t^2 s_a/\sqrt{5} \approx 8.6\,\mathrm{mm}$ for this prior and sampling interval.

B Case Decomposition Details

The physics posterior (seed 8047, 40,000 proposals per stage) retains 22,583 airborne draws and has ESS 2,893. The illustrative draws have early-position RMSE below 5 mm (indices 6953, 2827 and 18017). The responsibility of component k at $\mathbf{q}$ is proportional to $\pi_k \mathcal{N}(\mathbf{q}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$.

Weighted correlations are -0.911 between ω_y and Z, $+0.908$ between ω_z and Y, and -0.046 between C_D and Y. The force depends on $\boldsymbol{\omega} \times \mathbf{u}$, which couples spin components. Among draws with $Y < 2.45\,\mathrm{m}$, 58% of posterior weight has $|\omega_z| > 55\,\mathrm{rad\,s^{-1}}$. The spin prior and lift cap are both plausible contributors to the asymmetry.

At fixed $Y = 2.62\,\mathrm{m}$, seed 11 has slice maxima at $Z = 1.86$ and $2.16\,\mathrm{m}$. The three marginal Z densities have single maxima at 1.89, 2.05 and 2.04 m. The physics slice is sensitive to smoothing, which limits inference about distinct flight regimes.